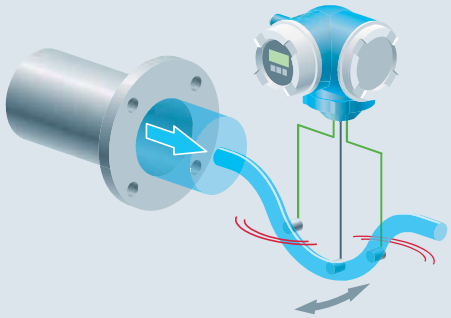
Coriolis mass flowmeters

Proline Promass – Multivariable sensors and highest accuracy: just two of the many reasons why the Coriolis measuring principle is being used more and more frequently to measure gases and liquids



Measuring principle

Each Coriolis flowmeter has one or more measuring tubes which an exciter causes to oscillate artificially. As soon as the fluid starts to flow in the measuring tube, additional twisting is imposed on this oscillation due to the fluid's inertia.
Two sensors detect this change of the tube oscillation in time and space as phase difference. This difference is a direct measure of the mass flow. In addition, the fluid density can be determined from the oscillation frequency of the measuring tubes.
The temperature of the measuring tube is also registered to compensate thermal influences. The process temperature derived from this is available as an additional output signal.

Advantages at a glance

- Universal measuring principle for liquids and gases
- Multivariable – simultaneous measuring of mass flow, density, temperature and viscosity
- High measuring accuracy:
  – typically: ±0.1% o.r.
  – optionally: ±0.05% o.r. (PremiumCal)
- Measuring principle independent of the physical fluid properties and the flow profile
- No inlet/outlet runs necessary

Measuring principle movie:
www.eh.digital/3jr2hUv

Simultaneous measurement of mass flow, density and temperature opens up entirely new perspectives for process control, quality assurance and plant safety. Additional important characteristic values can also be calculated from the primary variables measured:

- Volume flow and standard volume flow
- Solid content and concentration of binary mixtures: e.g., °Brix, °Plato, %mass/vol
- Characteristic variables in the petroleum industry: e.g. net oil, water cut, °API, weighted averages

The Coriolis measuring principle is used in a wide range of various industrial branches, such as the life sciences, chemicals, petrochemicals, oil and gas, food and – no less importantly – in custody transfer applications. Virtually all fluids can be measured: cleaning agents, solvents, fuels, crude oil, vegetable oils, animal fats, latex, silicon oils, alcohol, fruit solutions, toothpaste, vinegar, ketchup, mayonnaise, gases or liquefied gases.

Over 1 million Endress+Hauser Coriolis flowmeters have been successfully installed since 1986.







Promass sensors

Promass F
For universal use



- Tube material: stainless steel, Alloy C22
- Up to –196 °C (–321 °F)/ +350 °C (+662 °F)
- For custody transfer (MI-005, cryogenic fluids, MI-002)
- DN 8 to 250 (⅜ to 10")

F L **E** X

Promass A
For low flows



- Tube material: stainless steel, Alloy C22
- Self-drainable single-tube design
- DN 1 to 4 (1/24 to 5/32") (up to 430.9 bar/6250 psi)

F L **E** X

Promass Q
The top specialist



- Highest performance in density and custody transfer applications
- Unmatched accuracy in foaming liquids with "Multi-Frequency Technology"
- DN 25 to 250 (1 to 10")

F L E **X**

Promass P
Life sciences industry



- Hygienic single-tube system
- Complies with ASME BPE, ISPE, FDA, EHEDG and 3-A
- Tube material: stainless steel
- DN 8 to 50 (⅜ to 2")

F L E **X**

Promass E
For basic applications



- Cost-effective sensor
- Tube material: stainless steel
- DN 8 to 80 (⅜ to 3")

F L **E** X

Cubemass C
For low flows



- Space-saving, multivariable measuring system
- Tube material: stainless steel
- DN 1 to 6 (1/24 to ¼")

F L **E** X

Promass G
Compact, high pressures



- Ultracompact design with threaded connections
- Up to 350 bar (5076 psi)
- Tube material: stainless steel
- DN 8 to 25 (⅜ to 1")

F L E **X**

Promass H
For aggressive fluids



- Single-tube system
- Tube material: zirconium, tantalum
- Highest corrosion resistance
- DN 8 to 50 (⅜ to 2")

F L E **X**

Promass X
Maximum flow rates



- For oil and gas
- Highly accurate four-tube measuring technology
- Tubes and housing: stainless steel
- For custody transfer
- DN 300 to 400 (12 to 16") (up to 4100 t/h)

F L E **X**

Promass K
For utilities



- Cost-effective general-purpose device
- Minimized total cost of ownership
- Optimum usability via SmartBlue app or touch screen
- DN 8 to 80 (⅜ to 3")

F **L** E X

Promass I
Straight single-tube



- Easy-to-clean single-tube system
- Tube material: titanium
- Optionally with viscosity measurement
- DN 8 to 80 (⅜ to 3")

F L E **X**

Promass S
Food industry



- Hygienic single-tube system
- Standard approvals: 3-A, EHEDG and FDA
- Tube material: stainless steel
- DN 8 to 50 (⅜ to 2")

F L E **X**

Promass O
For highest pressures



- For oil and gas
- Corrosion resistant measuring tubes made of Super Duplex
- Stainless steel housing
- For custody transfer
- DN 80 to 150 (3 to 6") (PN 250)

F L E **X**

Proline transmitters

Proline – the perfect transmitter for every application
Whether applied in the life sciences, water, or food industry, the new generation of Proline transmitters can be freely combined with many of our sensors, all tried and tested for decades – while maintaining full backwards compatibility. This means a high level of added value and maximum flexibility matching your industry and application demands.

Detailed description of all Proline transmitters ▸ page 9 to 11

Promass sensors	Proline transmitters				
	10	100	200	300	500
					
Promass F	–	✔	✔	✔	✔
Promass E	–	✔	✔	✔	✔
Promass K	✔	–	–	–	–
Promass A	–	✔	✔	✔	✔
Cubemass C	–	✔	–	✔	✔
Promass I	–	✔	–	✔	✔
Promass Q	–	–	–	✔	✔
Promass G	–	✔	–	–	–
Promass S	–	✔	–	✔	✔
Promass P	–	✔	–	✔	✔
Promass H	–	✔	–	✔	✔
Promass O	–	✔	–	✔	✔
Promass X	–	–	–	✔	✔